

MINI PROJECT USING JAVA.

Assignment no - 02

Problem statement -

Write a program in Java showing hierarchical inheritance with base class as Employee & derived classes as FullTime Employee & InternEmployee with methods Display Salary in base class and Calculate Salary in derived class.

Calculate salary method will calculate as per increment given to fulltime and intern Employee. Fulltime employee - 50% hike. Intern employee - 25% hike. Display salary before and after hike.

Objective -

1. To study inheritance in Java
2. To study why to use inheritance
3. To study types of inheritance.

Theory -

1) What is inheritance in Java

Is an object oriented concept in Java where one class (subclass / derived class) acquires the properties & method of another class (superclass / base class). It is implemented using the extends keyword.

Eg -

FullTimeEmployee & InternEmployee inherit common properties from the Employee class.

2. Why to use Inheritance ?

- Promote code reusability

- Reduce cost duplication
- Improve maintainability
- Support method overriding
- Represent IS-A relationship.

3. Types of Inheritance in Java.

- Single inheritance - one subclass inherits one superclass.
- Multilevel Inheritance - chain of inheritance.
- Hierarchical Inheritance - multiple subclasses inherit one superclass.
- Multiple Inheritance - not supported using classes, achieved using interfaces.

Algorithm -

1. Start
2. Create a base class Employee with salary data member
3. Define method display salary() in base class.
4. Create derived class FullTimeEmployee.
 - Calculate salary with 50% hike
5. Create derived class InternEmployee.
 - Calculate salary with 25% hike
6. Display salary before & after hike
7. Stop.

Class diagrams.

